

Consequence-Aware Failure Diagnostics for Learned Particle-Reconstruction Representations

Collaboration concept note

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Abstract

A monitor must answer two questions that average reconstruction accuracy does not: what declared mechanism caused an alarm, and what scientific consequence followed. We propose a replay-based diagnosis of frozen representations under controlled interventions. The central contrast separates acquisition-contract violations from clean rare-subgroup physics; the corresponding mechanisms are movement along task-relevant versus redundant representation directions and loss of support in learned neighbourhoods. We test whether alarms attribute these families and whether their magnitude predicts downstream loss, explicitly measuring high-alarm/low-consequence and low-alarm/high-consequence cells. The testbeds are public neutrino-telescope point clouds and controlled waveforms: a public TIDMAD denoising benchmark plus a simulator in which assumed and realized noise covariance are known. Layerwise evidence is compared against input tests, MMD, classifier two-sample tests, embedding distance, uncertainty, and output-only monitoring at a matched false-alert budget, with abstention on families the study never declared.

1 Gap and contribution

Existing shift monitors can detect changes in inputs, outputs, or embeddings, but a scientific reconstruction pipeline requires two additional distinctions: whether the change reflects corrupted acquisition or valid but under-supported physics, and whether the detected change materially degrades a physics result. We test whether intermediate representations improve this attribution beyond generic monitoring at the same false-alert budget, while allowing abstention on ambiguous or unseen failures. The attribution claim is conditional on the declared simulator and forward model: unknown physics and model misspecification are not identifiable from a response profile alone and are routed to abstention.

In waveform reconstruction, structured acquisition noise induces a likelihood geometry through the inverse covariance: the correct residual weighting is Σ^{-1} , and a model evaluated under the wrong weighting can prefer reconstructions that are worse in the metric the instrument implies. Architecture and training coverage further determine which physically relevant distinctions survive into the representation. This study asks whether violations of these assumptions can be diagnosed after training, and whether a detected violation is scientifically consequential. The artifact is a reproducible experiment suite—configurations, adapters, hooks, replay scripts, and reports—not a maintained software product.

Identifiability limit. Diagonalizing a covariance yields decorrelated statistical modes, not labeled physical sources:

$\Sigma = \Sigma_1 + \Sigma_2$ admits infinitely many positive-semidefinite splits, and low-rank factors are rotation-degenerate. All attribution here is over *declared intervention families* under a stated simulator. Observing a response profile does not identify the cause of an unknown real shift, and we make no such claim.

2 Failure contracts

Three declared labels, frozen before evaluation.

N—acquisition-contract violation. For this study the closed family is power-spectral-density (PSD) or covariance change, jitter, drift, spectral lines, glitches, tails, hit thinning, and channel or module loss. The measurement violates a declared acquisition condition; no broader N claim is made.

S—training-support shift. Window or event composition moves toward physical strata that are sparse under the clean training distribution. The events themselves are clean and physically valid; only the support has moved.

E—evaluation-contract fault. Wrong units, coordinate ordering, output semantics, PSD convention, or metric implementation. This is a pipeline bug, not a distribution shift.

N versus S is the scientific core. E is decided by deterministic schema and range invariants, so it is reported as a gate result, not as a class in a classification score. The informative and difficult separation is N from S. N and S may both move the observed input distribution, so generic input-shift detection alone cannot distinguish them. Evidence for N comes from acquisition-quality indicators, paired corruption interventions, or known violations of the measurement model. S is defined by physically valid, uncorrupted events that occupy low-density regions of the clean training support. The scientific problem is therefore not whether the input changed, but whether the change reflects corrupted measurement or valid but under-supported physics. That distinction is what matters operationally: discarding S rejects physically valid events, whereas rejecting N removes measurements that violate the declared acquisition contract.

3 Claims

C1—Detection. At a 1% false-alert rate on nominal, non-overlapping windows, the best monitoring signature detects at least 80% of held-out abrupt and gradual N/S shifts within five windows, and is non-inferior to the strongest eligible baseline by a five-point power margin. Delay is primary in *windows*.

C2—Attribution with abstention. Two endpoint families are co-primary. The operational family is macro-F1 over $\{N, S\}$ on held-out cases, with N+S mixtures scored

by multi-label F1. The mechanism family is standardized displacement, neighbour retention, subspace principal angle, and the layerwise sensitivity profile, each normalized by clean within-stratum dispersion.

A split-conformal nonconformity score [4] is used to *abstain* on cases that do not resemble the declared training families. Unseen-family detection is evaluated by AUROC, while attribution quality under abstention is reported through selective risk and macro-F1 as functions of retained coverage [5]. The selective endpoint is risk-coverage AUC, with N/S macro-F1 at 80% retained coverage reported alongside it. Detection of the unknown and attribution among the known are thus not scored as if they were the same quantity.

The paired difference $\Delta F1$ between the full layerwise and all-generic arms is reported with a 95% confidence interval, and read three ways. A practically meaningful benefit requires the interval to exclude zero *and* the point estimate to reach 0.10 macro-F1; equivalence is supported if the interval lies within a predeclared ± 0.05 margin; otherwise the result is inconclusive. This lets a strong generic baseline produce a meaningful negative result — internal hooks may not justify their cost. A mechanism metric is promoted from development to the frozen confirmatory set only if it responds monotonically to the positive-control severity and adds separation beyond the generic arm; otherwise it remains descriptive.

C3—Cost. Ten-percent event sampling plus fixed-size sketches yields p95 replay-path latency overhead $< 10\%$, $\geq 10\times$ less representation storage, and ≤ 5 points of power or F1 loss relative to full capture.

C4—Is an alarm informative about scientific consequence? Let A be the monitor’s alarm magnitude and K the downstream scientific loss for the same intervention cell. High consequence is $K \geq \kappa_m$, where κ_m is a testbed-specific scientific tolerance frozen before evaluation — provisionally a 10% degradation for the waveform amplitude task, and for angular or clustering metrics a physically meaningful change in resolution, efficiency, purity, or ARI rather than a generic percentage.

Table 1: C4 alarm-consequence matrix and endpoints.

	$K < \kappa_m$	$K \geq \kappa_m$
Low alarm	quiet/benign	missed consequence
Strong alarm	harmless alarm	detected consequence
Endpoints	Primary: among strong-alarm cells, AUROC/AUPRC for $K \geq \kappa_m$ and conditional high-consequence risk; also the low-alarm/high-consequence rate. Secondary: within-family $\rho(A, K)$ and the high-alarm/low-consequence rate.	

The waveform testbed decides this, because the generator supplies both the assumed $\hat{\Sigma}$ used by the encoder and the realized Σ , permitting a controlled test of whether alarms correspond to degradation in the correct likelihood-weighted residual. C4 is primary there and secondary on the point-cloud testbed, where consequence is downstream reconstruc-

tion error and no exact covariance exists. We state that asymmetry rather than implying symmetric coverage.

4 Testbeds and monitoring design

Neutrino-telescope point clouds (realism). One primary public configuration with a released checkpoint exposing a 128-dimensional pre-head event representation. Module dropout, hit thinning, and time jitter instantiate N; S windows use clean low-density strata in energy, topology, vertex/containment, and multiplicity, estimated from training data only. One secondary architecture serves as a consistency check. A paired detector-geometry contrast is the first joint pilot because both views can share the same latent event and only physically supported acquisition changes; the arm remains gated on successful event pairing and matching.

Waveforms (mechanism; parallel track). The synthetic arm uses a noise-aware linear control and a generator that constructs PSD/cross-spectral density models, samples seeded frequency-domain coefficients with real-signal symmetry, and adds declared artifacts. The public TIDMAD arm uses a $\sim 0.6M$ -parameter two-stage domain transformer and the release’s denoising score, comparing otherwise identical MSE and inverse-PSD-weighted objectives. This compact transformer is a diagnostic subject, not a foundation model; a larger pretrained model is only a later gated comparison.

Monitoring comparison. Each window records input-quality summaries, named-layer moments or sketches, outputs, uncertainty, delayed task metrics, and calibration. A regularized logistic classifier is fitted with identical splits to five arms: input only; output/uncertainty; final embedding only; all generic-monitor outputs; and the full layerwise signature. The fourth is the honest strong baseline, committed to in advance. Eligible baselines — corrected univariate input tests, RBF-MMD [6], a classifier two-sample test [7], embedding mean/covariance distance, output tests, uncertainty/entropy — receive identical windows and false-alert budgets. Clean miscalibration is a model result, not automatically a software bug [8]; latent displacement is normalized by clean within-class dispersion. Held-out severities, seeds, support strata, mixtures, and one unseen family prevent perturbation lookup; windows and thresholds are frozen after development; intervals resample event groups and perturbation seeds separately; null results require a passed positive control, adequate precision, and a predeclared equivalence margin.

Table 2: Core experiment matrix. C1–C3 run on both testbeds; C4 is primary on waveforms.

ID	Test	Primary endpoints
E0	Clean replay; deliberate E faults	Reference discrepancy; gate sensitivity/specificity
E1	Held-out abrupt/-gradual N/S	Power at 1% false alert; AUROC/AUPRC; delay in windows
E2	N/S attribution, mixtures, unseen family	Macro/multi-label F1 over {N,S}; paired C1 vs generic arm; unknown AUROC
E3	Capture vs sample/sketch	p50/p95 latency; bytes/event; power/F1 loss
E4	Alarm vs consequence (C4)	$\rho(A, K)$; AUROC for $K \geq \kappa_m$; off-diagonal rates

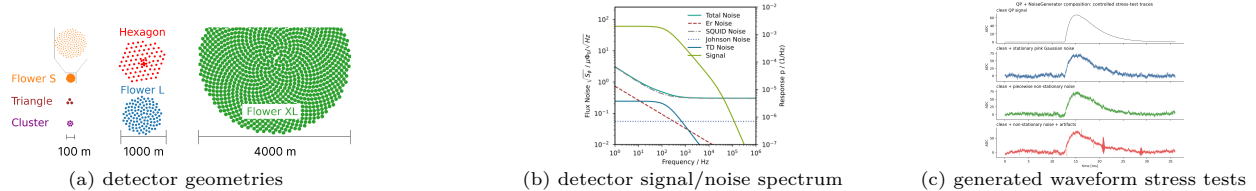


Figure 1: Complementary testbeds. (a) Six top-view geometries, reproduced from NuBench Fig. 3 [1]. (b) The supplied $\text{Al}_2\text{O}_3/\text{Al}$ athermal-detector response and component PSDs. (c) The generator composes a clean optimal-filter-like pulse with stationary colored noise, piecewise nonstationarity, and explicit artifacts. Panels (b)–(c) are project outputs.

5 Staged joint roadmap

Phase 1 (B): paired detector geometry. Run the first joint pilot on common latent events under one declared acquisition intervention. Without the pairing gate, report only a consistency comparison, not a causal layout claim.

Phase 2 (A): SPINE/Graph-SPICE transfer. On one frozen checkpoint, test one physically justified perturbation and one point-representation metric against clustering efficiency, purity, fragmentation, or merging near vertices.

Phase 3 (A $\oplus$ B): target experiment. If both pilots pass, compare physics-supported geometry/acquisition interventions with matched non-physical representation changes at the same displacement magnitude. This is the unifying attribution test, not an assumption that every learned coordinate is meaningful.

Relation to chained-model uncertainty propagation. Douglas et al. [3] propagate input uncertainty through a chained neutrino-LAr reconstruction and compare uncertainty-aware and blinded models under synthetic noise. Here the model is frozen: the question is whether its representation attributes a declared failure, abstains outside scope, and links an alarm to consequence. Propagated uncertainty is therefore a candidate covariate, not the endpoint.

6 Collaboration

The agreed sequence is $B \rightarrow A \rightarrow A \oplus B$, while the waveform/TIDMAD arm runs in parallel and supplies the first code-complete contrast. Work should move to a new shared repository rather than debugging the old inaccessible link. A proposed fortnightly meeting reviews one frozen protocol or result; initial ownership is waveform infrastructure and generic monitoring on Dowling’s side, and geometry/SPINE physics choices and validation on Junjie’s side, with thresholds and claim changes approved jointly. The first meeting should freeze one geometry intervention, pairing variables,

one representation endpoint, and one downstream physics endpoint.

7 Feasibility gate

Dataset access, checkpoint loading, released-prediction rescoring, intermediate-hook extraction, and a monotone module-dropout positive control have been validated ($15.28^\circ \rightarrow 25.12^\circ$ median angular error and 10-NN retention $1.000 \rightarrow 0.393$ across 0–50% dropout). Exact re-inference parity remains unresolved: the current pipeline differs from the released directions by 0.944° median angular error. Exact identity is a software control, separate from the scientific geometry gate. On a fixed development subset we will test whether that offset is stable across severity and event strata. Stable offsets permit a predeclared backend tolerance for the paired contrast; severity-dependent drift is a confound and keeps the gate closed.

References

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